

# Portfolio



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Selected Work · August 2026

# Table of Contents

- [ServiceMob · Lead AI Software Engineer, "Mob AI"](#)
- [TrueConsent · Build YC's Next Unicorn \(2nd of 1,168\)](#)
- [ISIS Neutron and Muon Source · Predicting Reliability with Machine Learning](#)
- [RE:JOIN · Software Engineer, Full Stack](#)
- [NLU · Claim-Evidence Detection](#)
- [Cognitive Robotics · CNN for CIFAR-100](#)

## ServiceMob • Lead AI Software Engineer, "Mob AI"

Irvine, California • Oct 2025 – present • Lead AI Software Engineer, leads 2 engineers

Started at [ServiceMob](#) as an AI Software Engineer Intern. The CTO left during early production of the company's flagship product, a multi-tenant B2B SaaS contact-center intelligence platform. I inherited an unfamiliar stack with a hard deadline, turned fuzzy asks into a concrete MVP, and designed the system end-to-end as sole engineer at launch. I was promoted from intern to full-time and, as of July 2026, am the **Lead AI Software Engineer**, leading two engineers.

### Production AI agent, "Mob AI"

- Built on Anthropic Claude (Sonnet, Haiku, Opus) via AWS Bedrock and the [Model Context Protocol](#) (MCP).
- Wrote **two custom MCP servers from scratch** (ClickHouse analytics, call-transcript retrieval) with closure-bound tenant isolation enforced server-side, tool routing, streaming and structured outputs.
- Re-architected the agent loop off the Claude Agent SDK onto direct Anthropic Messages API calls, cutting **time-to-first-token from ~20s to ~1s** (a 20x improvement).

### RAG pipeline

- Built a production **RAG pipeline** (Vertex AI embeddings, Qdrant vector store, hybrid retrieval) grounding the coaching agent in real resolved cases for **1,000+ users**; batched Athena ingest cut scan cost **87×**.

### CI-gated LLM evaluation harness

- Golden fixtures with frozen manifests; hybrid scoring **50% deterministic tool-use checks + 50% LLM-as-judge** for correctness, completeness and faithfulness; blocks any deploy regressing more than 10% from baseline.
- Drove the production model decision, showing Sonnet and Haiku matched Opus accuracy at **1/14th the cost** (a ~93% inference-cost reduction), with CloudWatch SLO monitoring (p50/p95), tracing and spend alarms.

### Dynamic SQL builder pipeline

- Typed, multi-tenant query-composition framework in TypeScript (3 packages, 67 internal consumers) with a node-based DSL, dependency-graph traversal and CTE hoisting; one registry powers **62 metrics across 173 analytics endpoints**.

**Outcome:** shipped the MVP **2 weeks ahead of schedule**; pilots converted to retained paying clients; platform scaled from **200 to 1,000+ active users**; ~185K LOC across both repos.

## TrueConsent • Build YC's Next Unicorn (2nd of 1,168)

Build YC's Next Unicorn • Agent Hack Day • Apr 2026 • Lead of 3 (Advit, Jwalin Shah, Victoria Wang)

A Y Combinator-affiliated build competition, **1,168 participants**, ending in a live demo to a YC panel. We took on US insurance denials: 1 in 5 claims denied, under 1% ever appealed, denials written in CPT/ICD-10/CARC codes designed to make people give up.

**What we built:** TrueConsent, a full-stack app that turns any scanned Explanation of Benefits, denial letter, itemized bill or prior-auth denial into a complete, grounded appeal packet in under 60 seconds. That pipeline is the fully-built core: we looked every code up against public CMS tables locally before the model touched the document, constrained the LLM to a single structured tool call, ran a deterministic rule engine, and verified every surfaced flag verbatim against its source. On top of it, our live demo placed the outbound call to the insurer through a Bland AI voice agent that read the appeal into the IVR and returned a reference number on a live transcript.



*Receiving the Build YC's Next Unicorn award, 2nd of 1,168*

**Outcome:** 2nd of 1,168 participants, prizes from Lightsprint and Hackersquad, project continuing post-hackathon.

# ISIS Neutron and Muon Source • Predicting Reliability with Machine Learning

University of Manchester × [UKRI](#) • Sept 2023 – Apr 2024 • BSc dissertation • First Class (82%)

The [ISIS Neutron and Muon Source](#) is a world-leading UK research facility whose particle accelerator runs continuously for weeks at a time. Downtime from ion-source failures disrupts live experiments across the whole facility. My dissertation partnered with UKRI to test whether **machine learning on real accelerator telemetry could predict those failures before they happened**, so maintenance could be scheduled rather than reacted to.

- Built the end-to-end pipeline in Python (Pandas + scikit-learn + Matplotlib): raw sensor logs to a time-window-expanded, labelled feature matrix.
- Ran EDA to prune the feature set (box-and-whisker plots, hierarchical clustering, KDE, PCA and a bespoke correlation heatmap), dropping redundant sensor channels.
- Tuned, trained and compared **five supervised algorithms** (Logistic Regression, Decision Trees, Random Forest, AdaBoost, XGBoost) across multiple time-window sizes, landing near **~90% failure-prediction accuracy**.
- Presented to UKRI through a written dissertation, a screencast and follow-up meetings; the work influenced UKRI's decision to keep investing, and I am writing it up as a peer-reviewed paper.

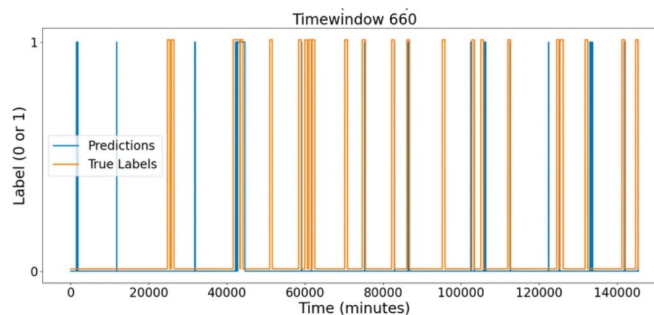


Figure 4.9: Prediction Visualisation of Logistic Regression - for time window expanded to 660 minutes.

| Time Window              | 660              |           |        |          |         |  |
|--------------------------|------------------|-----------|--------|----------|---------|--|
| MCC (3 d.p)              | 0.066            |           |        |          |         |  |
| Tabular Rectified Recall | 128335 (TN)      | 1006 (FP) |        |          |         |  |
|                          | 12 (FN)          | 10 (TP)   |        |          |         |  |
| Classification Report    |                  | Precision | Recall | F1-Score | Support |  |
|                          | 0                | 1.00      | 0.99   | 1.00     | 129341  |  |
|                          | 1                | 0.01      | 0.45   | 0.02     | 22      |  |
|                          | Accuracy         |           |        | 0.99     | 129363  |  |
|                          | Macro Average    | 0.5       | 0.72   | 0.51     | 129363  |  |
|                          | Weighted Average | 1.00      | 0.99   | 1.00     | 129363  |  |

Table 4.11: Summary Report for Logistic Regression with Time Window 660.

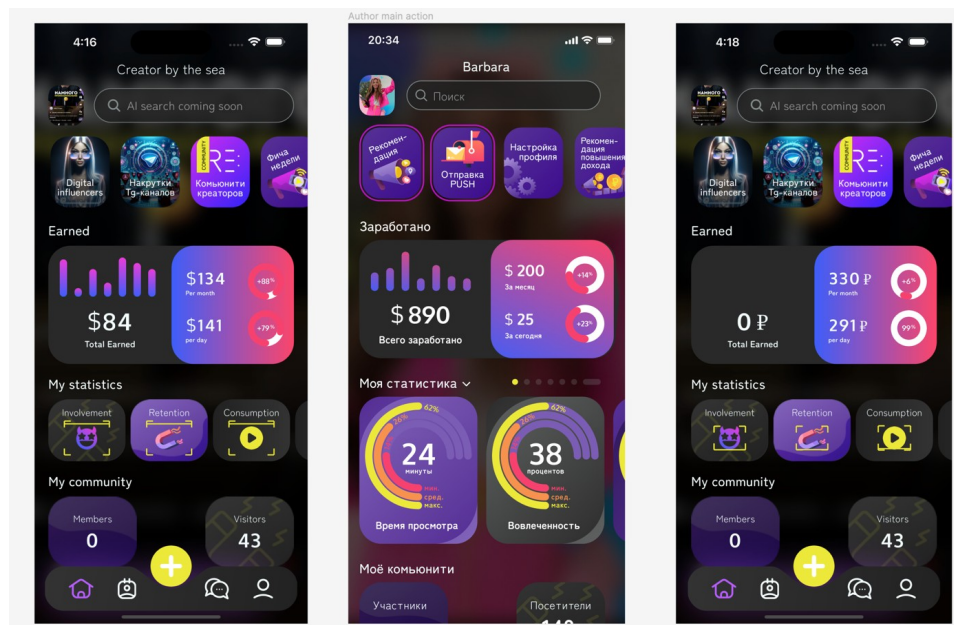
*Prediction visualisation (660-minute window) and Table 4.11 summary report*

## RE:JOIN • Software Engineer, Full Stack

Remote • Aug 2023 – Apr 2024 • part-time (concurrent with final year at Manchester)

Joined [RE:JOIN](#), a creator-economy platform, as a full-stack and mobile engineer on a small team (design, back-end and me).

- Shipped app revamps off Figma specs, resolving the top 3 reported multi-device bugs.
- Enhanced screens and widgets via dependency injection, lifting **shared UI reusability by 30%**.
- Engineered a **real-time chat room** between creators and clients, supporting **20+ active users** and lifting **engagement 40%**.



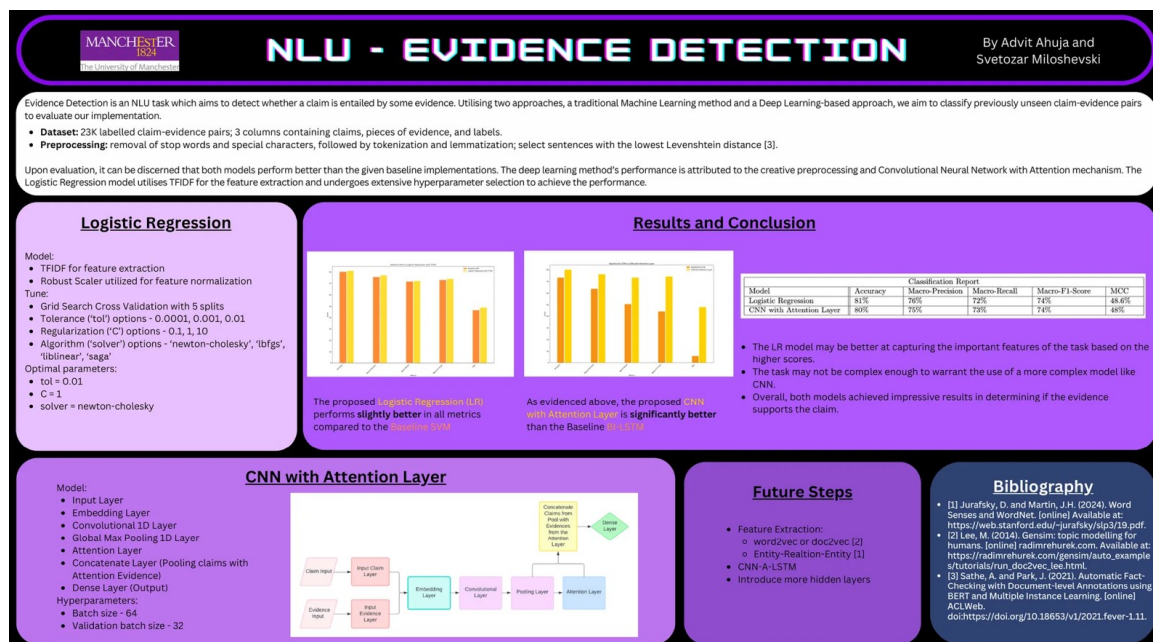
RE:JOIN creator dashboard and statistics

# NLU • Claim-Evidence Detection

University of Manchester • Mar-Apr 2024 • NLU coursework • team of 2 (with Svetozar Miloshevski)

Given 23,000 claim-evidence pairs, decide whether each piece of evidence supports its claim, comparing a classical ML approach against deep learning.

- Preprocessing: stop-word / special-character removal, tokenization, lemmatization; selected informative pairs by **minimum Levenshtein distance**.
- **Logistic Regression** over TF-IDF (Grid Search 5-fold CV) reached Accuracy 81%, Macro-F1 74%. A **CNN with Attention** reached Accuracy 80%, Macro-F1 76%, MCC 60%.
- Both beat the given baselines by **5-8%** on 5,927 unseen pairs; the simpler model generalized more efficiently at this data scale.



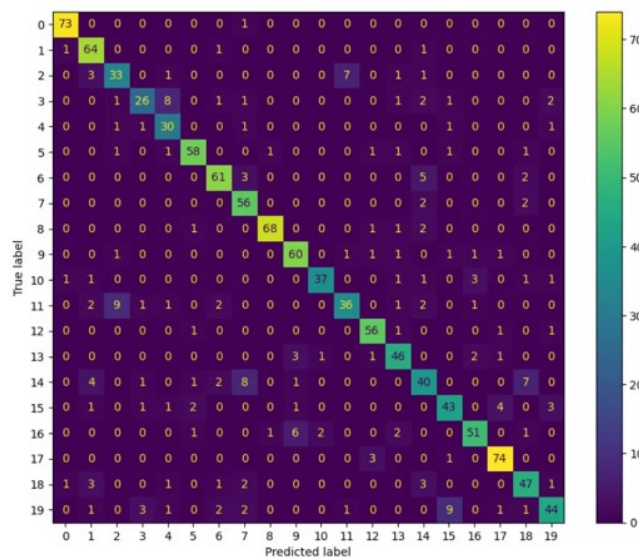
NLU Claim Evidence Detection poster, results panel

# Cognitive Robotics • CNN for CIFAR-100

University of Manchester • March 2024 • Cognitive Robotics coursework • solo

Given a baseline CNN, tune it to classify CIFAR-100 images better.

- Modelled a CNN in TensorFlow / Colab; swept dropout, pool size, kernel size and learning rate, plotting accuracy-vs-epochs for each sweep.
- Best config reached test accuracy 0.54, a **+14% gain over the baseline** through systematic hyperparameter search.
- Confusion-matrix analysis exposed the model's failure modes (visually-similar class pairs), pointing at data augmentation as the next lever.



Confusion matrix, best configuration